

Databricks Infrastructure Cost Optimization Initiative

Q3-Q4 2025 Business Impact Report

\$229,536

Annual Savings Achieved

The Challenge

Infrastructure Costs Were Unsustainable

\$23,149/month

Peak Monthly Cost (August 2025)

Pool Clusters 24/7

Always-on compute
wasting resources

Over-Provisioned VMs

Mixed sizing with
40-60% idle capacity

Driver Bottleneck

Single driver causing
resource contention

No Auto-Shutdown

Clusters running
nights & weekends

The Solution

3-Phase Optimization Strategy

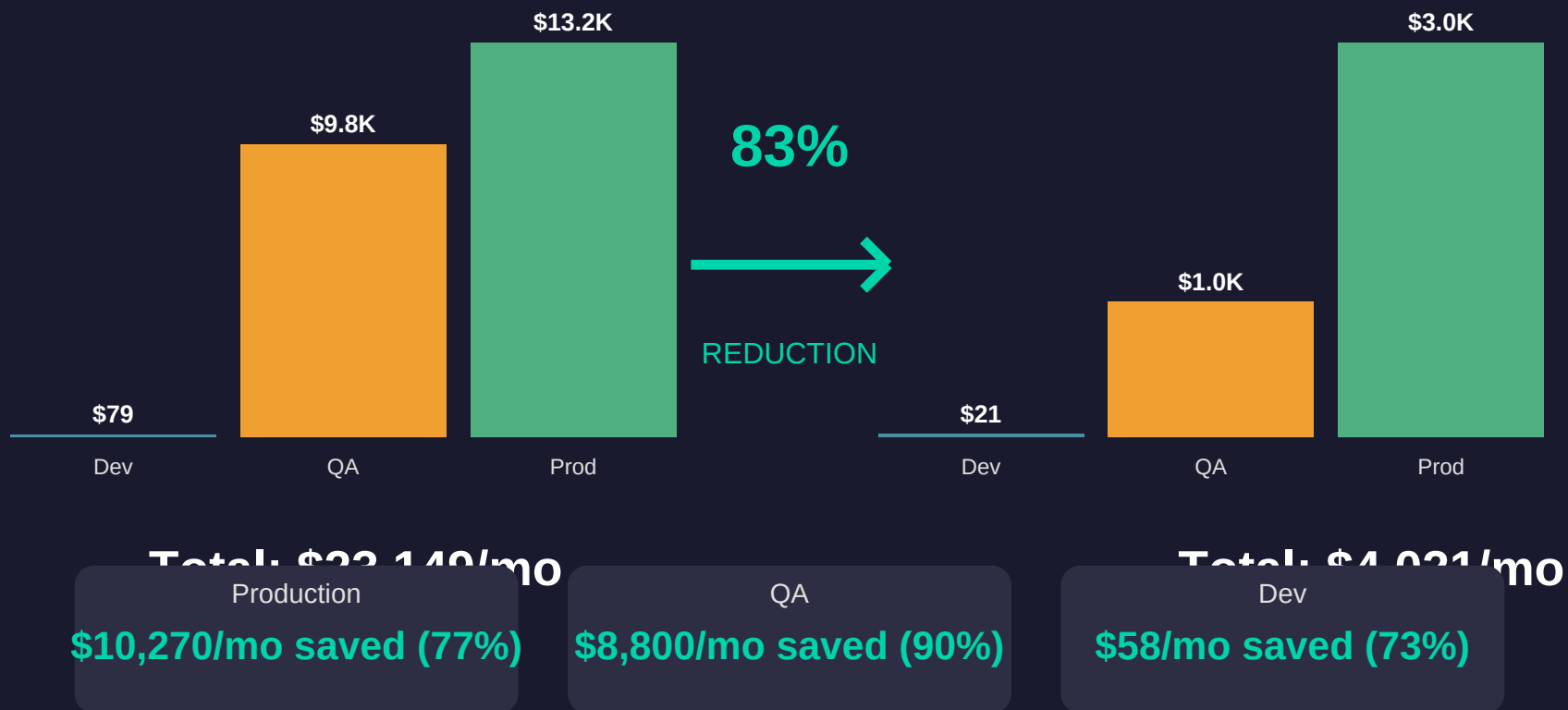


Cost Reduction by Environment

Before vs After Optimization

BEFORE (August 2025)

AFTER (November 2025)



Monthly Cost Trend

6-Month Optimization Journey



Additional Optimizations

Comprehensive Infrastructure Improvements

Photon Engine

2-3x faster queries
Native vectorized execution

Delta Tables

Z-ORDER clustering
Auto-Optimize enabled

Smart Scheduling

Off-peak job execution
Dependency optimization

Storage Savings

Delta Lake compression
25% storage reduction

Monitoring

Cost alerts at 80%
Real-time dashboards

Governance

Cluster policies
Prevented over-provisioning

Results & ROI Summary

Measurable Business Impact

\$19,128

Monthly Savings

\$229,536

Annual Savings

83%

Cost Reduction

1,430%

ROI

BEFORE vs AFTER

Monthly Cost	\$23,149	→	\$4,021
Resource Utilization	35%	→	100%
Pipeline Duration	4.5 hours	→	1.8 hours
Weekly Failures	15 jobs	→	2 jobs

■ **Achievement: Cloud Cost Champion**

Data Engineering Team | Q3-Q4 2025